

The Purple-Pipe Loop

Ladera Ranch's runoff-recycling system: route resolved, treatment train documented, evidence assembled · LEHRP
· v3, 11 Sept 2026

Independent research project; not medical advice; establishes no causation. No contaminant is asserted present in any water or soil discussed; the relevant streams have not been publicly tested for the constituents examined, which is itself a central finding. Statement classes marked throughout.

1 · The loop is real — verified (A2)

District sources: it "collects urban return (runoff) water from Oso Creek, Dove Creek, and **Horno Creek** to **blend with the recycled water... for irrigation**"; the "**Horno Basin**... designed to capture, treat and deliver millions of gallons of runoff water **for local irrigation use**." Delivery is the purple-pipe (recycled, non-potable) system serving Ladera's slopes, parks, medians and common-area turf.

2 · The route, step by step — resolved from the permit record (A2)

Governing document: San Diego RWQCB **Addendum No. 4 to Order No. 97-52**, adopted 8 October 2008, permitting the **Horno Urban Water Recovery Facility** (Tract 15793, Horno Creek Hydrologic Subarea 901.27). Held in this project with SHA-256 hash.

1 — Streets to creek. Dry-weather runoff from roofs, lawns, streets and slopes enters the storm system and the Sienna Botanica vegetated corridor (~2.4 mi) and drains to Horno Creek.

2 — Creek to basin. Flow collects in the **Horno Water Quality Basin**, owned and operated by the **County of Orange**. Storm flows above capture capacity continue toward San Juan Creek and the ocean; the loop takes the low flows.

3 — Basin to wetlands. Diverted water passes "*through seven acres of constructed wetlands, owned by Ladera Ranch Homeowners Association*" — the entire engineered treatment train for this stream.

4 — Wetlands to the purple pipes. Pumped into the district's recycled distribution system: "*prior to blending with recycled water*" for "*subsequent landscape irrigation use*."

5 — Storage and pressure. The non-potable system is fed and pressurized from its storage (the ridge tanks). Exact alignment and blend point remain in the as-builts; the destination is documented.

6 — Sprinklers to soil, and around again. About **180 acre-feet (~59 million gallons) per dry season** sprayed on community landscapes; water evaporates or soaks in, and the next runoff returns down Horno Creek.

3 · Why the water does NOT filter through Chiquita — the proof (A2)

Proof 1 — the permitted sequence names no plant. The route is basin → wetlands → blend → irrigation. If runoff entered Chiquita's process it would *become* recycled water; instead the order consistently treats “diversion water” as a distinct stream that merely *blends with* recycled water.

Proof 2 — the monitoring provisions require two separate streams. Provision K.1: sample “*diversion water either prior to, or after mixing with, recycled water.*” Provision K.2: sample “*effluent from the recycled water facilities at the point of discharge, **prior to blending with diversion water.***” Two monitoring points for two streams meeting *downstream of the plant's discharge* — incoherent if the runoff had passed through the plant.

Proof 3 — the waived parameters are sewage-plant parameters. K.1 exempts the diversion water from BOD, Total Suspended Solids and Odor limits — classic wastewater metrics. The waiver exists because this stream is not plant effluent and was never processed as sewage.

Proof 4 — program design and geography. The district describes the sibling Upper Oso facility as collecting runoff and delivering water for irrigation — capture-and-deliver, not capture-and-treat-at-plant. The basin sits at Ladera's low point; the plant is across open space up Chiquita Canyon. The permitted design pumps to distribution, not across the canyon and back.

Net: the captured runoff receives **basin settling and wetland treatment only** — no tertiary treatment, no plant processes — before it is sprayed on the community. That is the permitted design and is typical for urban-runoff recovery; what makes it significant here is the untested legacy of the land it drains.

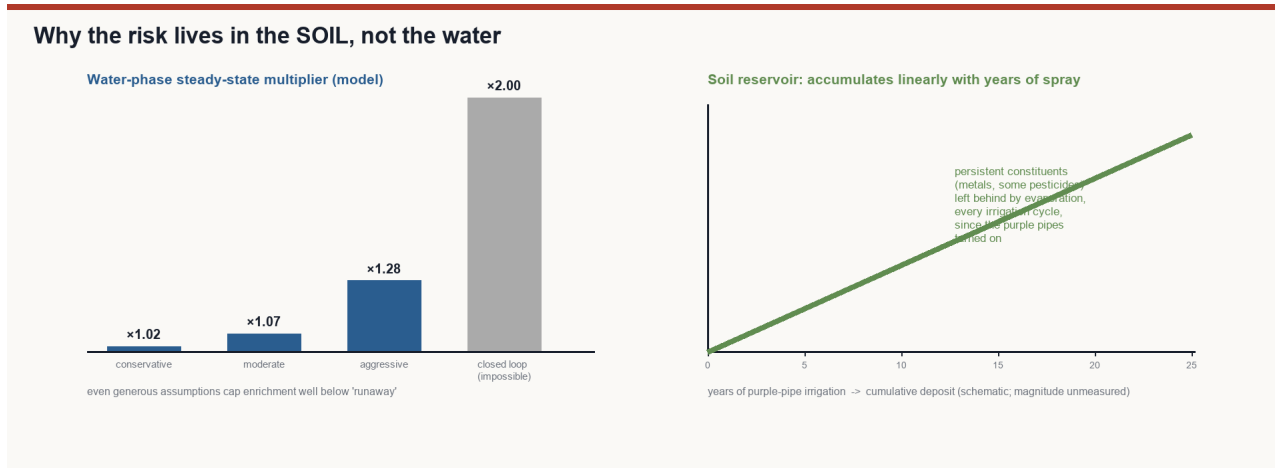


The resolved circuit on county 2025 imagery, numbered 1–6, with the Chiquita plant marked ruled-out for this stream. Pipe alignment schematic; destination and treatment documented.

4 · The audit — what fails, what survives

FAILS: water-phase “compounding.” The loop is partial (storm bypass, evapotranspiration, blending dilution): modeled enrichment caps at $\times 1.02$ – $\times 1.28$ even under aggressive assumptions.

SURVIVES: redistribution (drainage constituents re-sprayed community-wide); soil accumulation (evaporation leaves persistent constituents behind, linearly over two-plus decades); and the monitoring gap, now documented by the permit itself.



5 · Summary of evidence

#	Evidence	Source · grade	Establishes
E1	Runoff collected from Horno Creek “to blend with the recycled water... for irrigation”	SMWD page/FAQ · A2	The loop exists by design
E2	“Horno Basin... capture, treat and deliver... for local irrigation use”	SMWD On Tap 5/2012 (archived) · A2	Operator restates it publicly
E3	County basin → 7-ac HOA wetlands → blend into recycled distribution; ~180 AF/dry season	RWQCB Addendum 4 to Order 97-52, 10/8/2008 (held, SHA-256) · A2	Route and treatment train resolved
E4	Dual monitoring points; BOD/TSS/Odor waived for diversion water	Same order, Provision K · A2	Runoff and plant streams are separate — bypasses Chiquita
E5	Characterization sampling 11/2004–4/2005: TDS 686, sulfate 349, Na 57, Cl 48.5, B 0.22, NO3 0.6, F 0.275, P 0.4 mg/L	Same order, Finding 5 · A2	Salts and nutrients only — no arsenic, pesticide or pharmaceutical data
E6	CEQA Notice of Exemption, 9/29/2006	Same order, Finding 7 · A2	No environmental review of the diverted stream's chemistry
E7	Drinking water separate, imported, tested; arsenic ND–trace	District CCRs (held) · A1/A2	Tap water is not the issue; the non-potable loop is untested
E8	Recirculation mass balance: $\times 1.02$ – $\times 1.28$ cap	This report, transparent parameters · Model estimate	“Compounding” fails; redistribution and deposition stand

6 · Conclusions

C1 (verified). Ladera Ranch's dry-weather drainage is captured at the bottom of its watershed and returned to the community as landscape irrigation through the purple-pipe system — a designed, permitted, operating loop of roughly 59 million gallons per dry season.

C2 (verified). The returned water's entire treatment consists of a county-owned settling basin and seven acres of HOA-owned constructed wetlands. It does not pass through the Chiquita Water Reclamation Plant; tertiary treatment is not part of this stream's train.

C3 (verified records gap). Neither the permit's characterization sampling, nor its monitoring scheme as written in the addendum, nor any environmental review analyzed this stream for arsenic, pesticides or pharmaceuticals. Twenty years of operation; no public toxics data.

C4 (model + inference). The loop cannot make the water itself compound toxic — physics caps water-phase enrichment. Its significance is redistribution and deposition: what the drainage carries is sprayed community-wide, and persistent constituents accumulate in landscape soils.

C5 (hypothesis, testable). On land with a documented 1907–1912 arsenical dipping history and no soil testing, the loop is a plausible, unexamined redistribution pathway. Whether it carries anything of concern is unknown — and knowable for the cost of four samples.

The current data are insufficient to determine whether any constituent is present at levels of concern; the documented records gap and the land's history warrant the measurements below.

7 · What settles it

1) Irrigation water at a purple-pipe sprinkler head (arsenic + speciation, pesticide panel, CEC screen) — the cheapest, highest-value test. 2) Horno Basin sediment core. 3) Long-irrigated turf/slope soil vs never-irrigated control. 4) Wetland-outlet / blend-point water. Records worth obtaining: base Order 97-52 Table A-1 (is arsenic on the compliance list?); as-builts for exact alignment — three-party custody: County OCPW (basin), Ladera Ranch HOA (wetlands), SMWD (diversion works, Huitt-Zollars drawings).

Sources: RWQCB Addendum No. 4 to Order 97-52 (held, SHA-256 97479faa...d5771) · district recycled-water page and May 2012 newsletter (archived) · county aerial imagery 2025 · regional urban-runoff and water-recycling literature. LEHRP · corpus timestamped under version control.